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Query: Explain the Transformer architecture, including its key components like self-attention mechanisms, encoder-decoder structure, multi-head attention, and positional encoding, and its application in natural language processing tasks.

The Transformer is a type of model that significantly improves upon the standard sequence to sequence (seq2seq) models by introducing two novel aspects: self-attention mechanism and the use of encoder-decoder structure. These characteristics allow Transformers to process input data in parallel, allowing them to capture long-range dependencies without having to recalculate information at each step like LSTMs or GRUs.

Here are key components of a Transformer:

1. **Self-Attention Mechanism:** The self-attention mechanism allows the model to weigh the importance of words in a sentence when generating predictions for that sentence. In essence, it gives the model a global perspective on what's important to consider for each word within the sequence. A classic implementation involves three linear transformations of the input, resulting in queries, keys and values which are used to generate new representations.
2. **Encoder-Decoder Structure:** The encoder-decoder structure is typical of seq2seq models. It consists of two parts: an encoder that transforms the input data into a form understandable by the decoder; and a decoder, which takes the encoded information to generate predictions.
3. **Multi-Head Attention:** Rather than having a single attention head that attends to all positions in its input sequence, Transformers use multiple heads concurrently. These independent mechanisms each learn different things about the data, thus improving performance.
4. **Positional Encoding:** Unlike recurrent models like RNNs or transformers built from LSTMs/GRUs where words can be processed independently of their position in a sequence, Transformers include positional encoding to give them some understanding of the relative positions of words in a sentence.

Transformer applications in Natural Language Processing tasks:

1. Machine Translation: The transformer architecture has been shown to perform very well on tasks like machine translation where it's necessary to understand context across long sentences or documents, and is used by tools such as Google's BERT (Bidirectional Encoder Representations from Transformers) for machine translation and text summarization.
2. Question Answering: It can also be used in tasks like question answering where the goal is to find an answer from a given context, again showing its ability to process long documents with global information.
3. Text Summarization: Transformers have been very effective at text summarization task which involves condensing large pieces of texts into shorter ones. This reduces the cognitive load for readers and provides a quick summary instead of reading through entire documents.
4. Neural Text Generation: Transformer-based models are powerful tools to generate human-like text, particularly in tasks like creative writing or story generation where language has to be fluid and unpredictable.

In essence, the transformer is an innovative and efficient model for sequence processing, providing a substantial boost over previous seq2seq architectures such as LSTMs/GRUs. It's being widely used because of its superior performance on tasks like machine translation and question answering, while remaining computationally lightweight and requiring little memory overhead compared to previous models.